An Introduction to Mechanistic Interpretability

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REST-CL Retreat 🌴

May 30, 2026

Outline

Introduction

The Toolkit

Trends and Evolutions in the Field

Case Study

Why do Language Models output what they output?



- Output linguistically valid: correctly resolved pronouns, handled irony, processed metaphors, etc.
- The quality of the output can be measured quite efficiently

But what is happening between the input and the output?
Mechanistic interpretability tries to answer it.

Interpretability landscape

- **Behavioral**: analyzes input-output relations (black box)
- **Concept-based**: probes the model's learned representations for patterns and concepts
- Tell us *whether* the LM does something, not *how* or *why*.
- To analyze the underlying processes: mechanistic

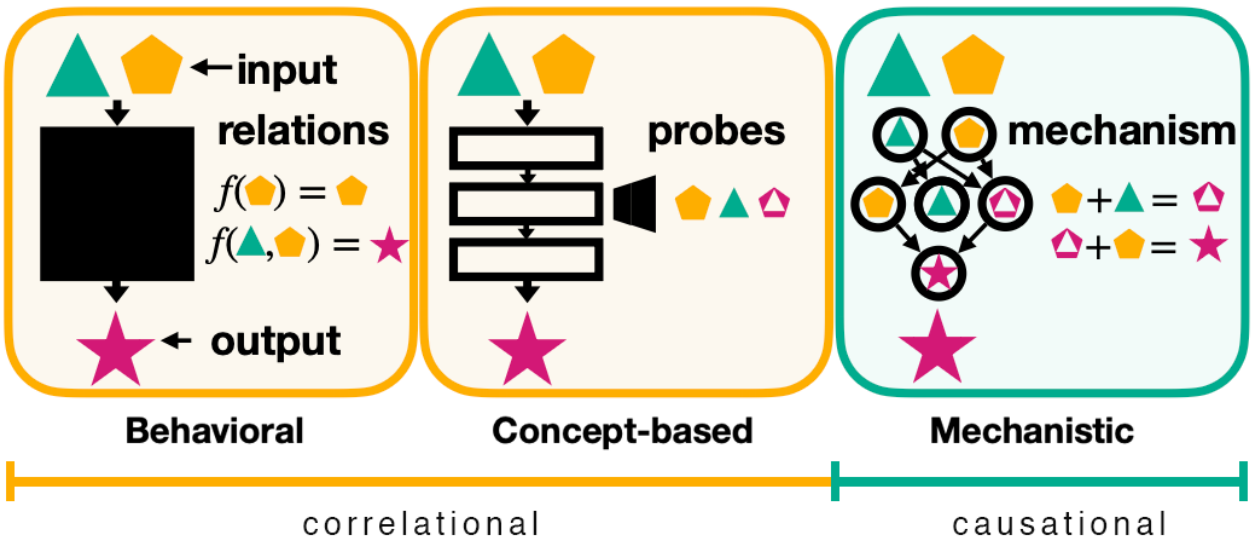
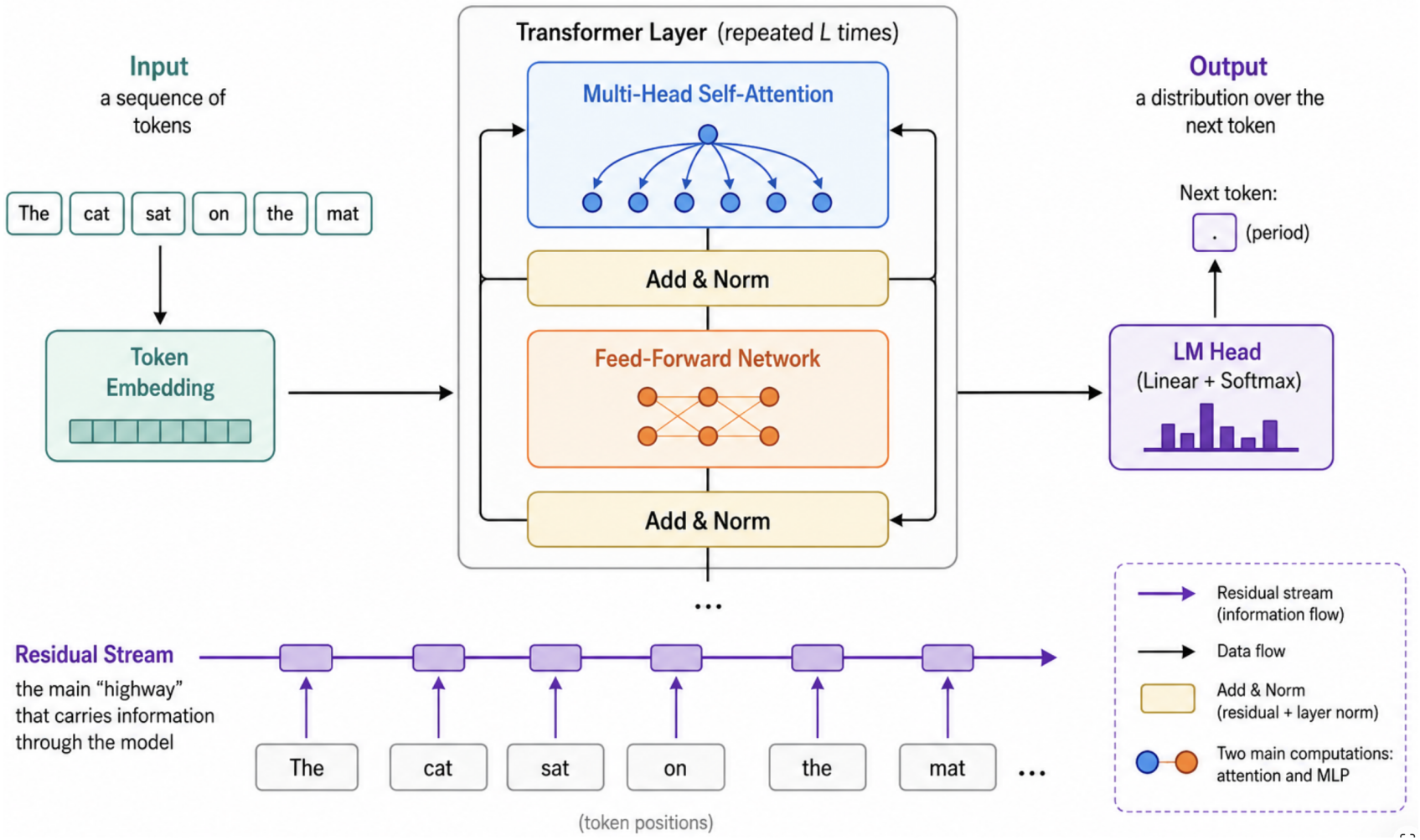


Figure: Adapted from Bereska & Gavves (2024)

Opening up a Transformer



Why is this difficult?

- **Distributed representation**: information is spread across many dimensions of a high-dimensional vector
- **Superposition**: the same neuron activates for multiple unrelated things

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Where to look?

In practice, you move between two levels:

Globally: the residual stream

- Investigate how the model's representation of the input is evolving
- Identify positions of interest: a specific neuron, head, layer

Locally: individual components

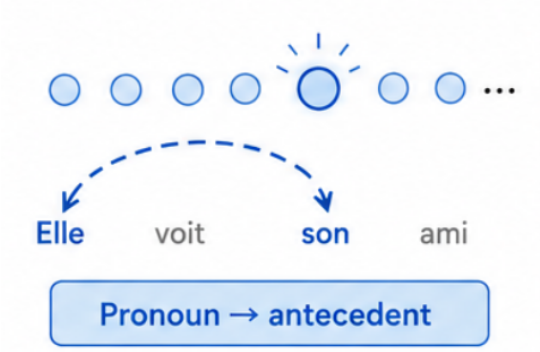
- Investigate what this component seem to compute

What can you find?

Things you can manipulate (in principle), in the form of:

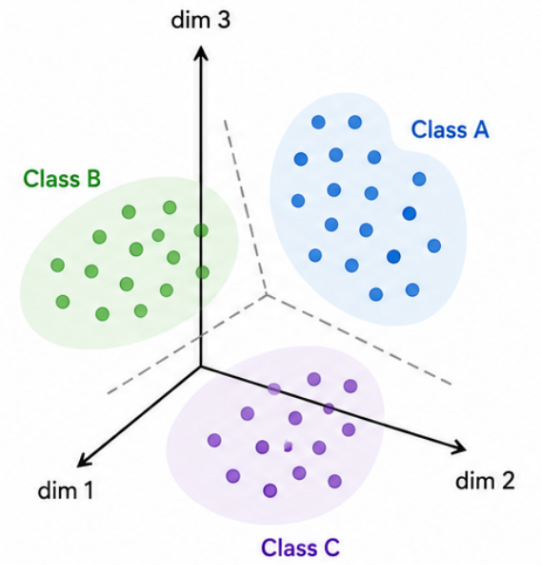
Features

Specific properties components appear to encode.



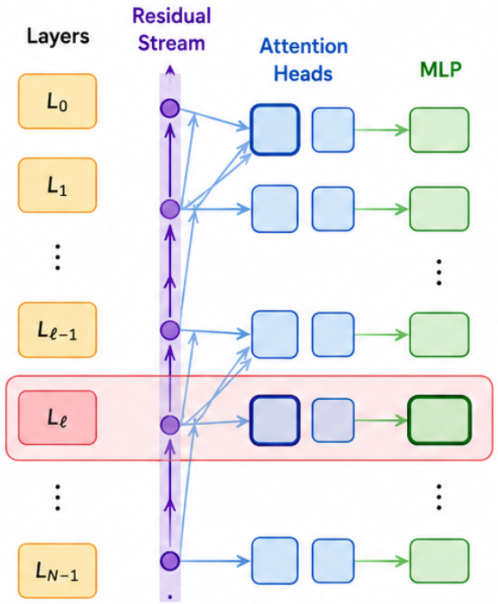
Representations

The geometry of activation vectors across layers and positions.



Circuits

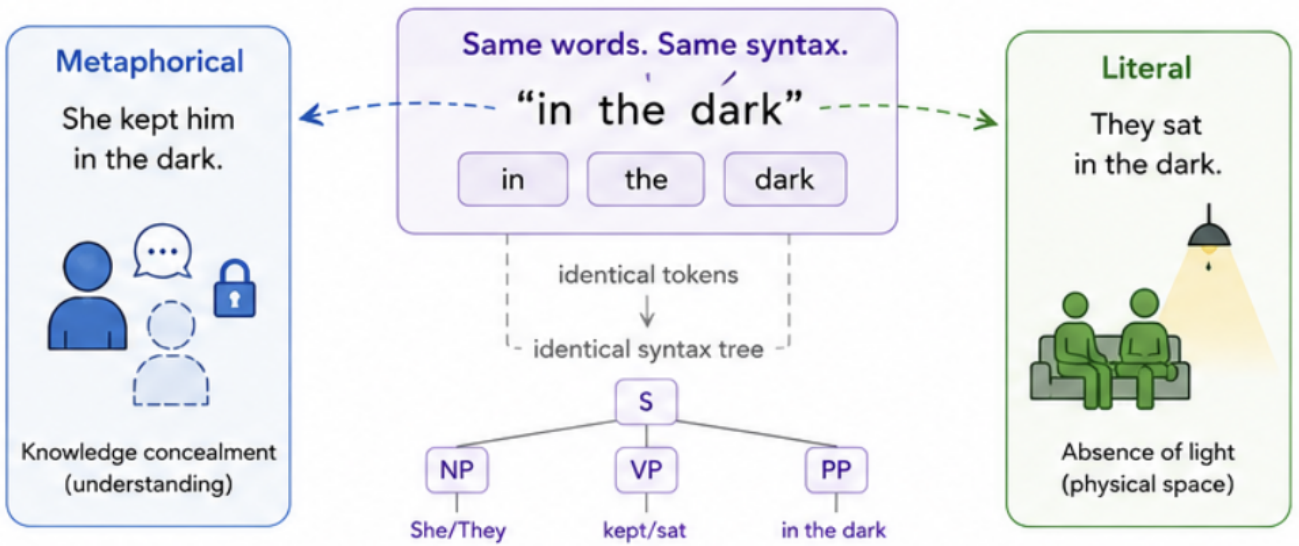
Subgraphs implementing a particular computation.



Use case: metaphor processing

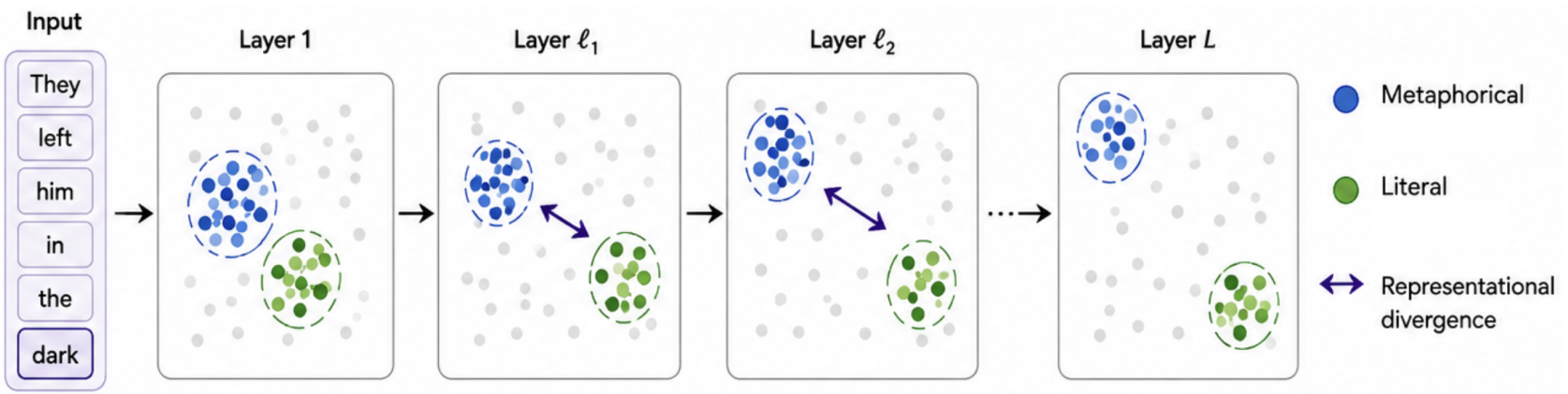
Question: when a language model processes a **metaphor**, does it compute something **structurally different** from when it processes the **same words literally**?

- Why metaphors? Abstract reasoning rather than pattern-completion.



Tool: Representational analysis

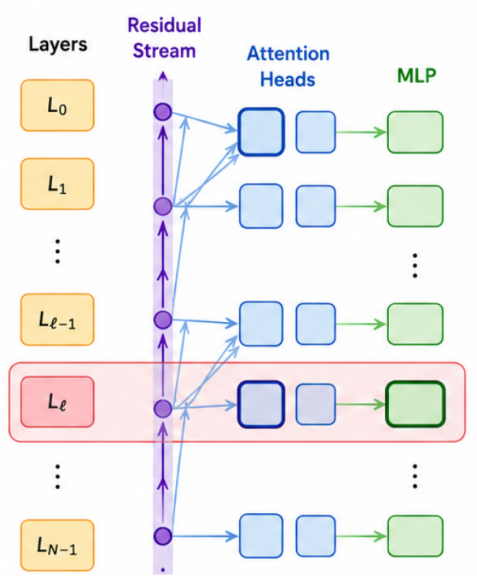
- Extract activation vector for a given token at a given layer
- Do this across many examples
- Do internal representation diverge?



Tool: Ablation (finding circuits)

Which components actually matter for the computation?

Goal: Identify the minimal set of components that are *causally sufficient* for a behavior.



Method

- Remove neurons, heads or layers
- Measure what behavior breaks
- Components whose removal changes the output are likely part of the circuit

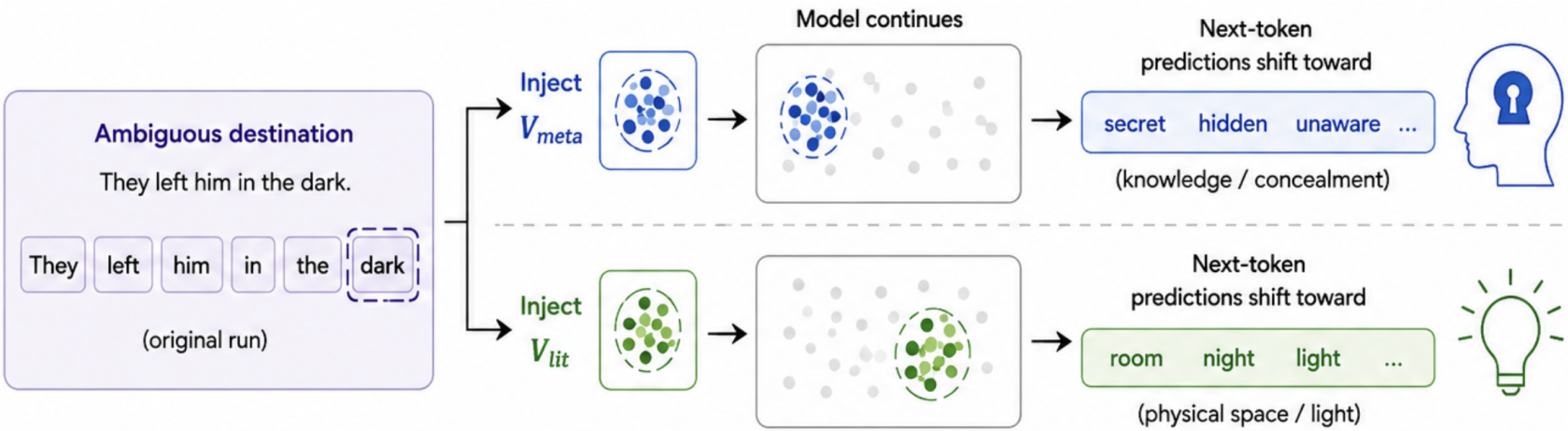
Ablation narrows the search from "the whole model" to a tractable computational subgraph

Tool: Activation patching

Intervene directly on the model's internal state.

- Extract the representation of *dark* from a metaphorical and a literal context
- Inject these representations into an ambiguous prompt
- Measure how next-token predictions change

Interpretation: Metaphorical processing if the injected representation systematically shifts the model toward unawareness; literal otherwise.



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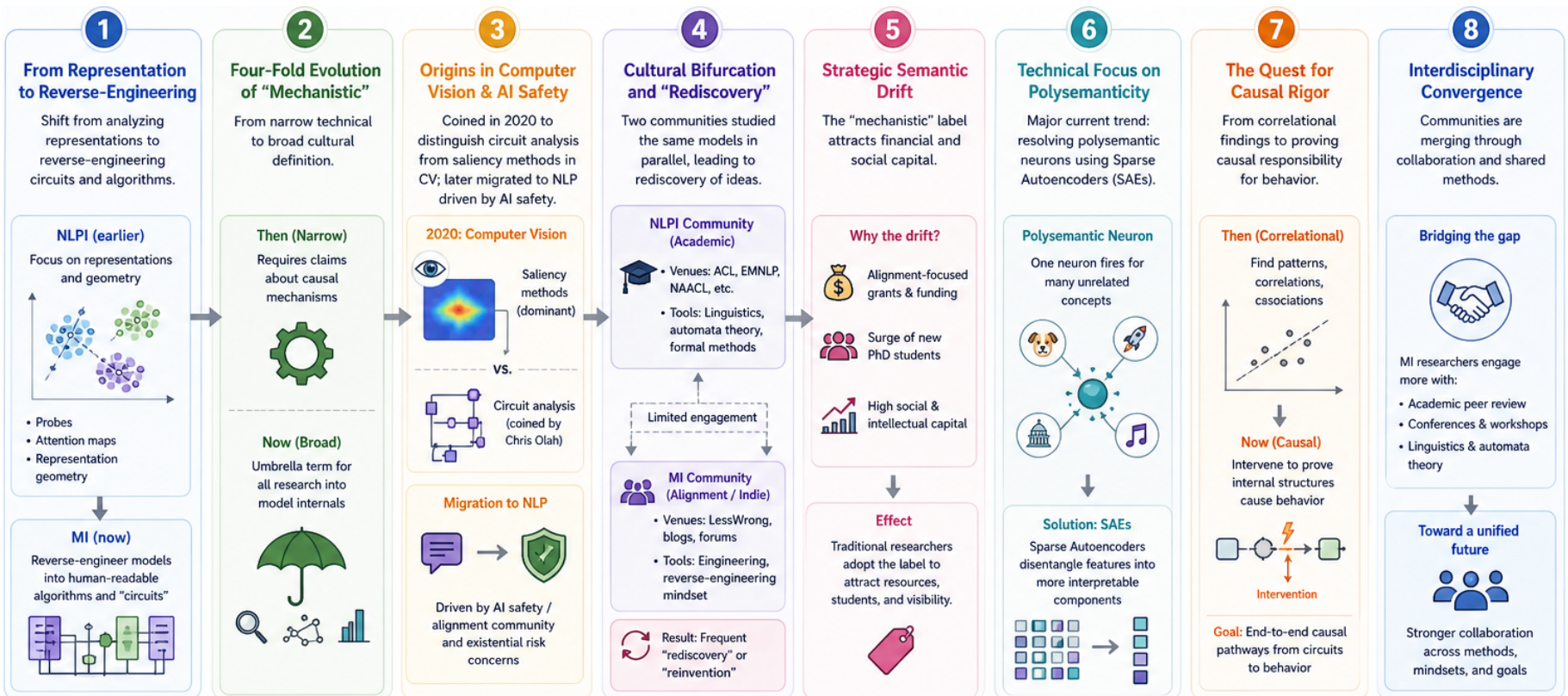
The Toolkit

Trends and Evolutions in the Field

Case Study

Trends and Evolutions

based on *Mechanistic?* by Saphra & Wiegreffe (2024)

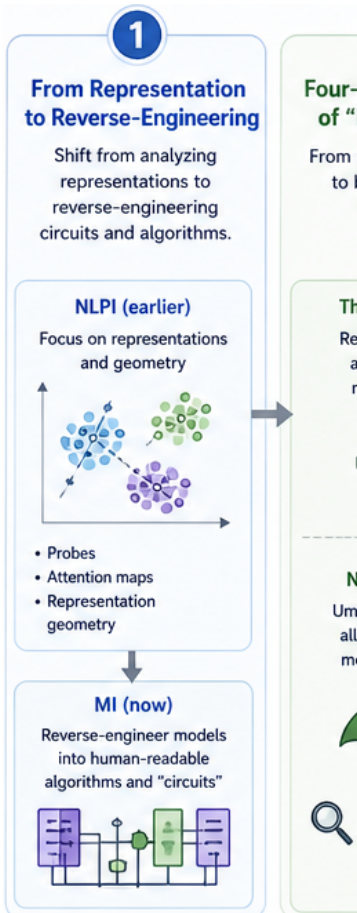


Trends and Evolutions

based on *Mechanistic?* by Saphra & Wiegreffe (2024)

1) Evolution from Representation to Reverse-Engineering

- CL/NLP interpretability earlier *(also connectionist work)*: linguisitic representations & geometric analysis
- MI: reverse-engineer e.g., LMs as human-readable algorithms & circuit-like structures

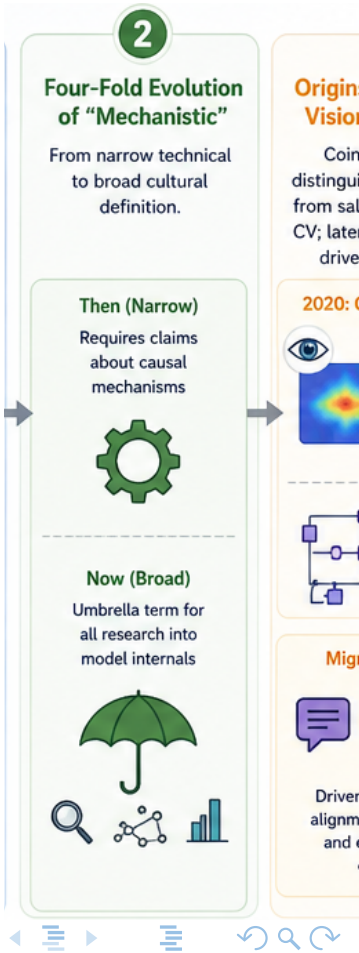


Trends and Evolutions

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2) The Four-Fold Definition of "Mechanistic"

- from a technical term
- to a cultural umbrella for investigating model internals (community driven shifts)

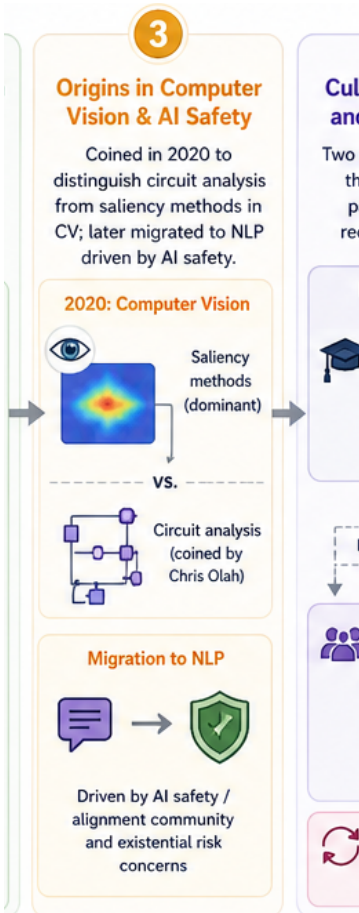


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3) Origins in Computer Vision and AI Safety

- Chris Olah (2020)'s blog post on circuit-based analysis baptized the field
- migration to NLP/Ai safety and alignment community

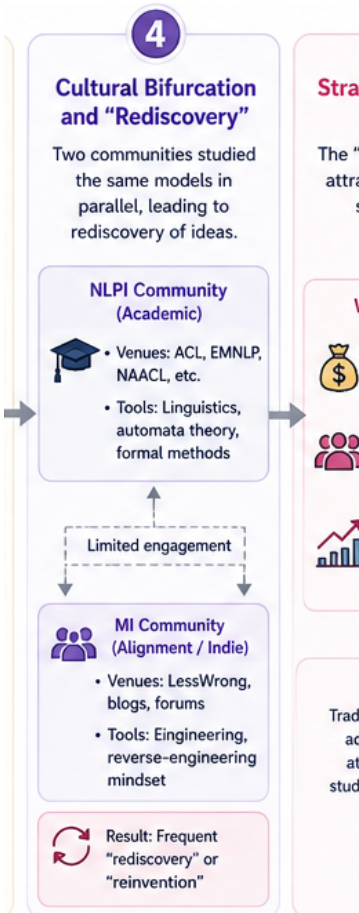


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4) Cultural Bifurcation and "Rediscovery"

- academic NLP community @ACL venues & co.
- MI community @blogposts *LessWrong*
- limited cross-communication: "rediscovery" of CL/NLP methods



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5) Strategic Semantic Drift

- umbrella term gains broader visibility and prestige
- funding from alignment focused organizations
- increased attention from prospective students



Trends and Evolutions

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6) Technical Focus on Polysemanticity

- a lot of work on world knowledge in LLMs with SAEs
- developing more interpretable methods to investigate internal activations

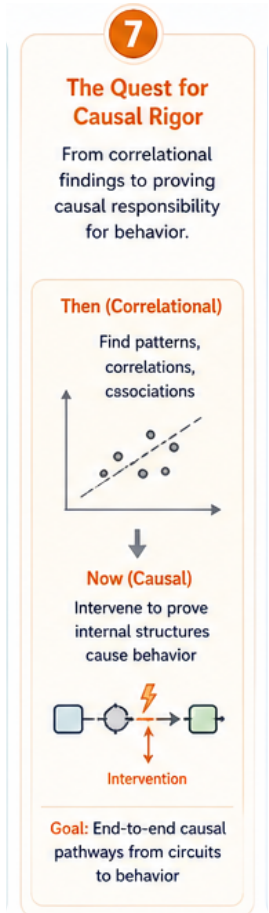


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7) The Quest for Causal Rigor

- probing-based methods critizized as highly correlational (e.g., Belinkov, 2022)
- e.g., Mueller et.al., 2025: *The Quest for the Right Mediator: Surveying Mechanistic Interpretability Through the Lens of Causal Mediation Analysis*



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8) Interdisciplinary Convergence

- MI researchers engage more with academic peer review
- incorporate tools from linguistics and cognitive science



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Case Study:
Interpretability Research on Syntactic Knowledge in Language Models

The Grammar of Transformers:
A Systematic Review of Interpretability Research on Syntactic Knowledge in Language Models

Nora Graichen[1], **Iria de-Dios-Flores**[1*], **Gemma Boleda**[1,2*]
Universitat Pompeu Fabra[1], ICREA[2]
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- systematic review of 337 papers/ about 3000 datapoints from a range of syntactic phenomena and interpretability methods
- build a quantitative, transparent evidence database beyond prior narrative reviews

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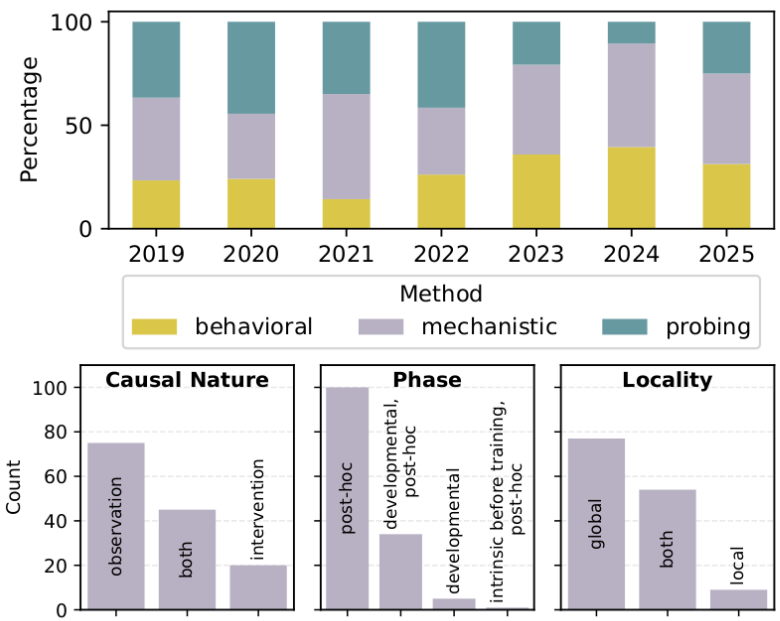
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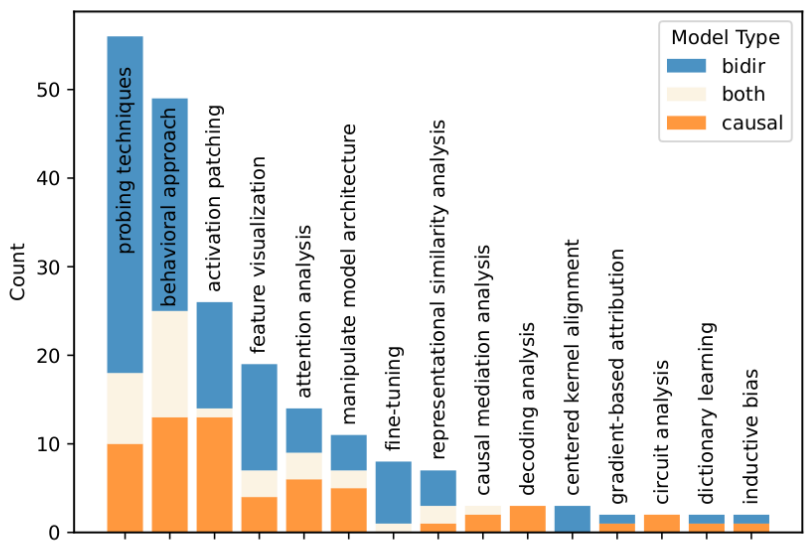
Nora Graichen[1], **Iria de-Dios-Flores**[1*], **Gemma Boleda**[1,2*]
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Interpretability Research on Syntactic Knowledge in Language Models

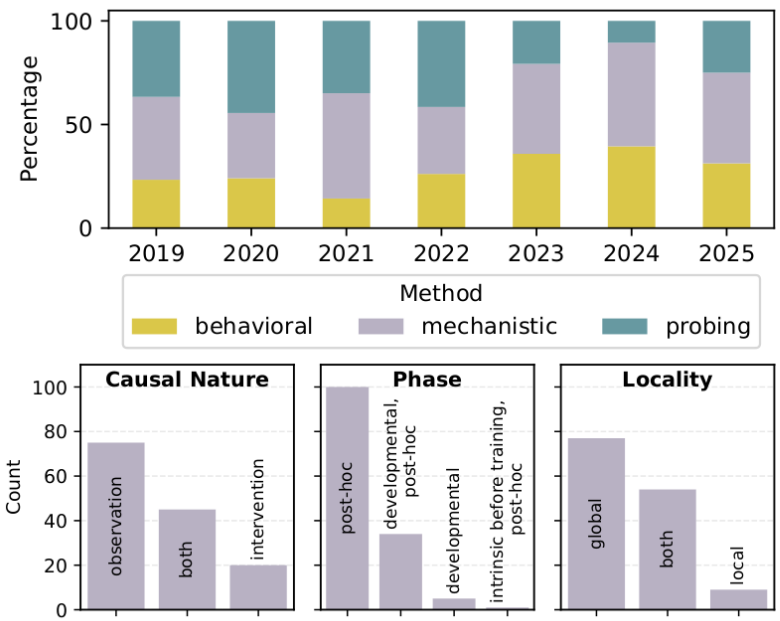
- goal: map research on syntactic knowledge in TLMs (*what has been done and how; what remains to be done*) and summarize current findings
- three paradigms coded for: behavioral, probing and mechanistic interpretability

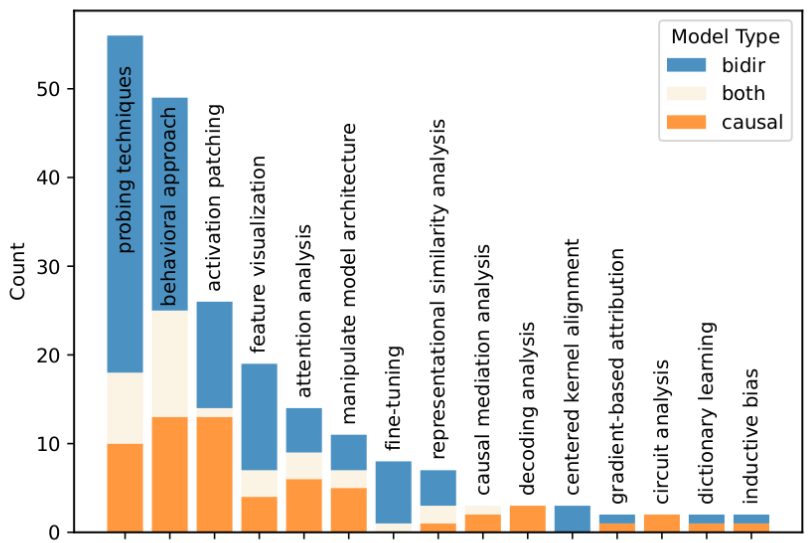




Interpretability Research on Syntactic Knowledge in Language Models

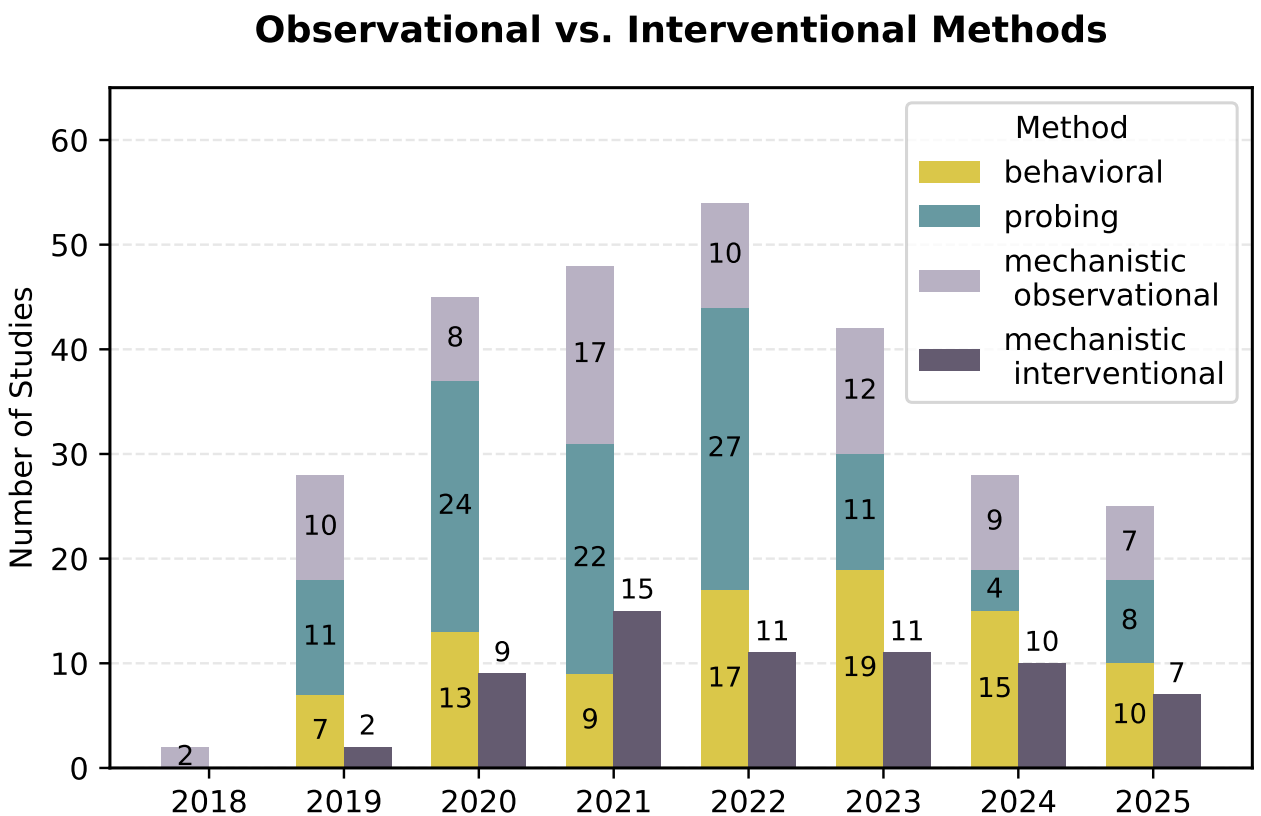
- analyzing trained models at a global level
- observational approaches dominate





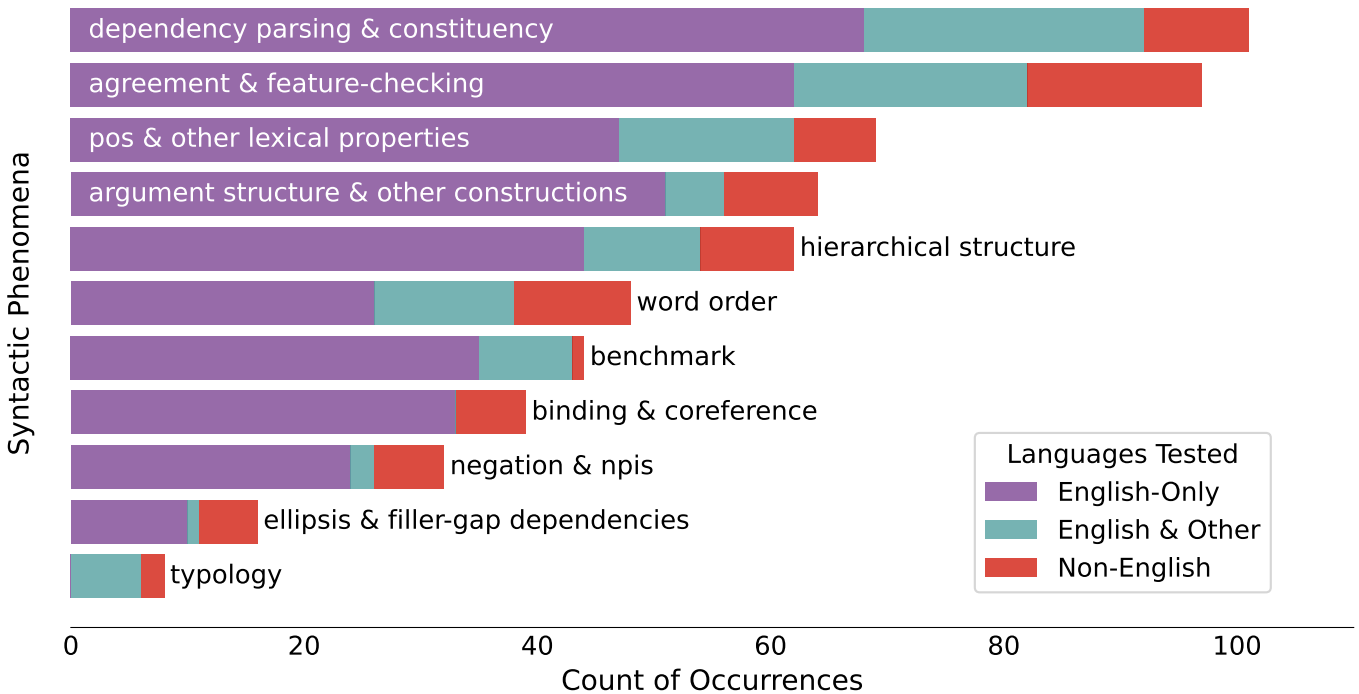
Interpretability Research on Syntactic Knowledge in Language Models

- interventional approaches still emerging
- diminishing trend of "only" probing-based studies



Interpretability Research on Syntactic Knowledge in Language Models

- major challenge: once studies are broken down by method and phenomenon to enable comparisons, resulting groups become too sparse and heterogeneous for a straightforward synthesis



Interpretability Research on Syntactic Knowledge in Language Models

- mechanistic approaches still emerging in the field
- no single method is sufficient in isolation
- strongest evidence comes from combining observational and interventional paradigms

Thank you! Any questions? 